rvLLM: Single-GPU, FP8, Graph-Captured LLM Inference in Rust

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https://github.com/m0at/rvllm

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Abstract

rvLLM is a single-GPU LLM inference engine written in Rust, built from scratch without Python in the serving hot path. On a single NVIDIA H100 SXM 80 GB running Qwen2.5-7B in FP8 E4M3 with 512 output tokens, greedy decoding, and CUDA graphs enabled, rvLLM reaches **19,287 tokens/second at** $N = 128$ (SHA `eb9e247fd`, 3 iterations, stdev 38.5).

The engine ships as a FlashAttention-3 SM90 paged-KV decode kernel (built from the Hopper source to `libfa3_kernels.so`), a CUTLASS 3.x SM90 FP8 GEMM variant catalog (40 non-residual + 10 residual-fused templates), a set of eight custom fused PTX kernels (norm+quantize, residual+norm+quantize, RoPE+KV-write, SiLU*mul+quantize, argmax, embedding, residual-add, post-attention quantize), and a graph-capture runtime that records **311 kernel launches per decode step into a single `cuGraphLaunch`** for each of 35 pre-captured batch buckets. Total device-to-host traffic per decode step is one $4 \cdot N$-byte copy of argmax'd token identifiers; total host-to-device traffic is one packed metadata buffer ($<$100 KB at $N = 128$).

The paper states the engine's five correctness invariants and enumerates the three structural bug classes that the engine is designed to eliminate: schedule-epilogue pairing mismatches in SM90 CUTLASS FP8 GEMMs (silent `CUDA_ERROR_ILLEGAL_ADDRESS` under graph replay only), captured metadata offsets clobbered by orthogonal upload paths, and stale paged-KV block tables from CoW-only change detection. Each is eliminated by a pinned variant, a frozen layout, or a typed signal from the scheduler. Missing build artifacts (autotune policy, FA3 `.so`, CUTLASS `.so`) refuse to start; no silent fallbacks exist in the FP8 dispatch. rvLLM is open-source under Apache-2.0.

1 Introduction

Inference serving for large language models is bandwidth-bound at decode batch sizes of practical interest. On an H100 running Qwen2.5-7B, a single decode step involves roughly 10–15 kernel launches per transformer layer, paged-KV attention over the sequence's current context, and a final projection to a 152,064-entry vocabulary. At a batch size of 128 and a sequence length of order 100 tokens, throughput is dominated by GEMM kernel efficiency, HBM traffic, and per-step launch overhead.

vLLM [1] established the paged-KV cache design and continuous batching scheduler that define the modern serving baseline. CUTLASS 3.x [**?**] and FlashAttention-3 [**?**] supplied the kernel primitives that make Hopper-specific WGMMA and TMA accessible to non-Python callers. rvLLM composes these primitives behind a Rust runtime that owns CUDA graph capture, paged KV allocation, metadata packing, and sampling DtoH coordination in 31 crates ($\sim$76K lines of Rust), with no Python interpreter in the serving path.

This paper documents the state of the engine at commit `eb9e247fd` (April 16, 2026), which achieves 19,287 tok/s at $N = 128$. Section 2 walks the full stack from HTTP handler to kernel. Section 3 enumerates the kernel inventory. Section 4 gives an ordered trace of one decode step. Section 5 states the five correctness invariants the engine enforces and ties each to a concrete failure mode observed during development. Section 6 describes three root-cause commits that,

on April 16, took the engine from 9,531 tok/s to 19,287 tok/s, none of which were stopgaps. Section 7 reports the full bench table; related work and limitations follow.

2 The stack, every layer

```
HTTP / OpenAI API          rvllm-api (axum)
Engine                     rvllm-v2::engine
                           step_pipelined() = launch + collect
                           double-buffered pinned argmax DtoH
Scheduler                  continuous batching
                           paged KV (block_size = 64)
                           page-boundary growth signal
Worker                     CUDA graph pool
                           35 pre-captured batch buckets
                           1 compute stream per worker
Runner                     1 HBM arena, pre-allocated slab
                           packed metadata (1 HtoD per step)
Layer                      11 kernel launches per layer
Kernels                    CUTLASS 3.x SM90 FP8 GEMM
                           FlashAttention-3 SM90 paged decode
                           custom fused PTX (norm / silu / rope)
                           cuBLAS HGEMM (LM head)
```

Figure 1: The rvLLM stack. Each row runs in a single Rust crate; the DAG is enforced by a compile-time test that parses every `Cargo.toml` and rejects out-of-layer dependencies.

2.1 HTTP, engine, scheduler

HTTP is an `axum` server exposing OpenAI-compatible `/v1/completions`, `/v1/chat/completions`, `/v1/embeddings`, and `/v1/responses`. Incoming requests are tokenized and placed on a lock-free MPSC queue consumed by a single worker thread; the worker is `!Send` by construction so a tokio runtime cannot accidentally schedule two concurrent GPU users.

The engine (`rvllm-v2::engine`) exposes two functions: `step_launch(diff)` enqueues all GPU work for one step (metadata HtoD, graph replay, argmax DtoH, event record) and returns immediately; `step_collect()` waits on the previous step's DtoH event and returns the new tokens. There is exactly one codepath; the synchronous variant that tied `cuStreamSynchronize` to each step has been removed.

The scheduler (`rvllm-scheduler`) implements continuous batching and paged-KV allocation. A request moves through `Queued` → `Prefilling` → `Decoding` → `Finished`. The scheduler emits one `BatchPlan` per step, of exactly one variant (`Prefill`, `Decode`, or `Idle`); prefill and decode never coexist in the same step. A watermark-triggered preemption path re-queues the lowest-priority decoding requests when KV-cache block use exceeds 96%.

2.2 Worker, runner, layer

The worker (`rvllm-worker` / `rvllm-v2::worker`) owns one CUDA compute stream, one CUDA context, and a pool of pre-captured graph instances indexed by bucket size. Buckets are $\{1, 2, 4, 8, 16, 24, 32, \ldots, 256\}$. Graph capture happens during engine initialization, not during warmup.

The runner (`rvllm-v2::runner`) owns one HBM arena allocated via one `cuMemAlloc` at startup, sized for the worst-case workspace (sweep over all CUTLASS variants $\times$ all bucket sizes $\times$ all GEMM shapes), plus the KV cache, scratch tensors, and packed metadata buffer. A compile-time `GraphSafe` trait marks tensors whose lifetime outlives the capture region; `&mut HbmArena` is not in scope inside a `CaptureScope` closure, so runtime realloc is unrepresentable.

The layer (`rvllm-v2::layer`) is a pure function of (input, weights, KV cache slab, scratch, metadata handle). One decode layer at FP8 is eleven kernel launches:

```
1.  fused_add_rmsnorm_fp8_quant
2.  CUTLASS FP8 GEMM (QKV)
3.  fused_rope_kv_write
4.  FA3 paged_decode
5.  quantize_fp8_per_token (post-attention)
6.  CUTLASS FP8 GEMM + residual (o_proj, v1)
7.  fused_rmsnorm_fp8_quant
8.  CUTLASS FP8 GEMM (gate_up)
9.  fused_silu_mul_fp8_quant
10. CUTLASS FP8 GEMM (down_proj)
11. residual_add_f16
```

Buffer aliasing is checked by the borrow checker: the layer takes `&mut Tensor<'a, f16>` for its output and `&mut LayerScratch` for scratch; two mutable borrows into one allocation fail to compile.

3 Kernel inventory

Every kernel has a pinned variant and a workspace contract. There are no dispatch fallback chains: each call site resolves to exactly one variant, determined at engine init from the autotune policy or pinned in source.

3.1 CUTLASS SM90 FP8 GEMM

CUTLASS 3.x templates drive the QKV, gate_up, down_proj, and fused o_proj+residual projections. Forty non-residual and ten residual-fused variants are instantiated, each parameterized by (TileShape, ClusterShape, KernelSchedule).

Variants cover tile shapes $\{64{\times}128, 128{\times}128, 128{\times}256\}$ in $M{\times}N$ with $K = 128$ or 256; cluster shapes $1{\times}1{\times}1$; and the `KernelTmaWarpSpecialized` (WS), `KernelTmaWarpSpecializedCooperative` (Coop), and their `FP8FastAccum` variants. Per-row activation scale and per-tensor weight scale are applied in the epilogue; the residual-fused kernel adds an `Sm90SrcFetch` fetch of the residual and a `TreeVisitor` computing $D = \alpha \cdot \mathrm{GEMM}(A, B) + C$.

Pairing constraint. The mainloop schedule must match the epilogue schedule. The residual-fused kernel uses `TmaWarpSpecializedCooperative` in its epilogue collective (`kernels/cutlass_fp8_gemm_res` line 100); any variant passing a non-cooperative mainloop with this epilogue causes `CUDA_ERROR_ILLEGAL_ADDRES` *only under graph replay* (the kernel runs cleanly outside a captured graph). The v2 runtime pins the o_proj residual path to variant `v1` ($128 \times 128 \times 128$ Coop/Coop matched); the v3 rewrite makes the mismatch a CUDA `static_assert`, rejecting the offending template pair at compile time.

Workspace contract. Each variant exposes `fp8_gemm_v<i>_workspace_size(M,N,K)`; the runtime's `max_workspace_size` query iterates all actually-callable variants across all bucket sizes for all GEMM shapes the model uses, and pre-allocates a single workspace slab accordingly. This eliminates the class of bug where the dispatch falls through to a variant whose workspace was never queried.

Autotune policy. A separate binary, `autotune-cutlass`, runs 67 HGEMM variants, 31 o-proj-residual variants, 32 gate-up-silu variants, 40 FP8 GEMM variants, and 10 FP8-GEMM-residual variants against every GEMM shape the model uses at every bucket size, reporting median of 10 timed runs after 3 warmups. The winning (shape $\rightarrow$ variant index) mapping is written to

`policy.json`, which is a build artifact shipped alongside the binaries. The runtime does not consult `~/.cache` for policy; stale caches from prior deploys cannot influence dispatch.

3.2 FlashAttention-3 SM90 paged decode

`libfa3_kernels.so` is built at deploy time from the FlashAttention-3 Hopper source tree plus an `extern "C"` wrapper (`kernels/fa3_sm90_wrapper.cu`). Build time is roughly 10 minutes on an H100; the resulting shared object is 23 MB. The wrapper exposes `paged_decode` and `paged_prefill` entry points. Both expect KV laid out as [2, num_blocks, block_size, num_kv_heads, head_dim in one contiguous `Region`; the factor-of-two comes from interleaving K and V per block.

GQA is supported via num_heads/num_kv_heads; `head_dim = 128` is a hard gate (the kernel is specialized and refuses other head dimensions). `context_lens[i] == 0` is a valid padded-slot marker and yields zero output without touching `block_tables[i,*]`.

No PTX fallback. A prior PTX-only attention kernel existed in the tree. It has been removed from the dispatch; if `libfa3_kernels.so` is not present at engine init, the engine refuses to load. The build script ships with the repository.

3.3 Fused pre/post kernels

Eight custom CUDA kernels compile to PTX and are loaded by the runtime. Each fuses one recognizable composite. No megakernels exist in the dispatch path.

Kernel	Fuses	Launches
`embedding_gather`	token_ids → f16 hidden	1 → 1
`fused_add_rmsnorm_fp8_quant`	residual + RMSNorm + quant	3 → 1
`fused_rmsnorm_fp8_quant`	RMSNorm + quant	2 → 1
`quantize_fp8_per_token`	post-attention act → FP8	1 → 1
`fused_rope_kv_write`	RoPE + paged KV write	3 → 1
`fused_silu_mul_fp8_quant`	SiLU + mul + quant	3 → 1
`argmax`	f32 logits → i32 token	1 → 1
`residual_add_f16`	$x + y$ in place	1 → 1

Each fused kernel ships with a pure-Rust f32 reference implementation in tests; the PTX must match within cosine 0.999 for FP8 outputs and 10^{-5} absolute for f32 outputs, over a fuzz of shape and value distributions. The `residual_add_f16` kernel is deliberately *not* fused into the GEMM epilogue; prior attempts to fuse residual with the downstream projection hit either schedule-pairing crashes or a ∼24% regression from increased tile pressure.

Vectorization rule. Kernels that accept dimensions divisible by eight halves issue `uint4` 128-bit loads (eight halves per load) and `uint2` 64-bit FP8 stores (eight FP8 per store), with register caching between the absolute-max reduction pass and the quantize pass so each HBM element is read once. Alignment is checked by a `debug_assert` in the Rust binding; misalignment fails the launch loudly rather than producing `CUDA_ERROR_MISALIGNED_ADDRESS` under graph replay.

4 One decode step, in order

For a decode batch of N sequences at bucket size $B = \text{pad_up}(N)$:

```
step_launch(diff):
  1. schedule(diff)                          CPU, O(N), zero allocs
  2. metadata pack + 1x HtoD                 packed i32 buffer:
                                             positions, context_lens,
```

```
                                          block_tables, slot_mapping
  3. graph_pool.replay(bucket = B)        one cuGraphLaunch:
       embedding_gather                    1 kernel
       for layer in 0..28:                 11 kernels x 28 = 308
         (as enumerated in Section 2.2)
       fused_residual_rmsnorm (final)      1 kernel
       cuBLAS HGEMM -> lm_head             1 kernel
       argmax                              1 kernel
  4. cuMemcpyDtoHAsync argmax -> pinned[w] 4*N bytes (512 B at N=128)
     cuEventRecord event[w]
  5. return; w ^= 1                       double-buffer flip

step_collect():
  1. cuEventSynchronize event[r]          waits for step-1 DtoH
  2. read pinned[r][0..N]                 CPU reads new tokens
  3. scheduler.commit()                   mark tokens, free finished
```

Total kernel launches per decode step: $1 + 28 \times 11 + 1 + 1 + 1 = \mathbf{311}$, all captured into **one** `cuGraphLaunch`. Total DtoH per step: one $4 \cdot N$ byte copy of argmax token identifiers. Total HtoD per step: one packed metadata buffer, <100 KB at $N = 128$ with max_blocks $= 129$.

The LM-head GEMM uses cuBLAS HGEMM (F16 in, F32 out) because CUTLASS FP8 autotune reported worse time on the $(N, 152064, 3584)$ shape at $M \geq 128$ than cuBLAS for the model tested. An FP8 LM-head path exists in `rvllm-v2` gated on the autotune cache's preferred variant; it is disabled by default for Qwen2.5-7B.

5 Correctness invariants

The engine enforces five invariants at build or startup time. Each is a response to a concrete failure mode observed during development.

5.1 No fallbacks

Rule. Missing autotune-policy entry for a shape panics with the shape. Missing `libfa3_kernels.so` refuses startup. Missing `libcutlass_kernels.so` refuses startup. The runtime does not read `~/.cache/rvllm/cutlass_autotune.json`; policy lives in the SHA-pinned `policy.json` shipped alongside the binary.

Failure mode this prevents. A prior deploy populated the on-box autotune cache with entries for an older CUTLASS catalog. A fresh build with the same HF-downloaded kernels inadvertently consulted the stale cache, resolving shape $(32, 152064, 3584)$ to a variant whose template no longer matched the `.so`'s symbol table, and ran the corresponding FP8FastAccum kernel under graph capture. Under graph replay, the kernel faulted with `CUDA_ERROR_ILLEGAL_ADDRESS`. SHA-pinning the policy artifact eliminates the failure class.

5.2 Single metadata-upload path

Rule. Metadata buffer layout is frozen per (bucket, max_blocks_per_seq). Captured graphs bind those exact device offsets. There is no non-padded upload variant. Prefill and decode use separate entry points that do not share offsets.

Failure mode. During development, two upload paths coexisted: `upload_metadata` (non-padded, sized to the actual request count) and `upload_metadata_padded` (sized to the captured bucket). On the first decode step after a prefill, the runtime's `last_meta_offsets` field carried

the non-padded offsets from the prefill call, but the captured graph still read from the padded offsets. The diff-upload path wrote new token IDs, positions, and block-table entries at the non-padded indices; the captured kernel interpreted the bytes at the padded indices as if they were current data, producing a `context_lens` value of order 10^5, a `total_tiles` of 780, and a `block_tables` read 391 entries past the slice end. The read resolved to an uninitialized region of the meta buffer, which held bytes of a small integer interpreted as `phys_block`. Multiplied by the per-block stride 16384, this yielded a device address 10.2 MB before the nearest valid allocation—verbatim the `compute-sanitizer` report of the original crash.

The v3 rewrite removes non-padded upload entirely. The v2 engine tracks `last_padded_batch: Option<usize>` and only takes the patch fast-path when it matches the requested bucket; otherwise a full padded upload happens.

5.3 Real block-change detection

Rule. The scheduler emits `ContinuedRequest::block_table_update: Option<Vec<BlockId>>` whenever a sequence's physical block list has grown since the last `mark_table_sent()`. The worker unions this with `diff.block_ops.copies` (copy-on-write events) to compute `has_block_changes` before each patch-decode upload.

Failure mode. An earlier version of the worker checked only `!diff.block_ops.copies.is_empty()`, which captures CoW events but *not* ordinary page-boundary growth. A sequence reaching the end of its current block and being allocated a new physical block generates no CoW event; the block-table entry at index $\lceil \text{context_len}/\text{block_size} \rceil - 1$ changes silently. The patch-decode path skipped the `block_tables` copy; the captured graph continued to read the stale block identifier, producing wrong KV-cache reads, garbage attention outputs, and garbage tokens. No crash. `ignore_eos=true` bench runs at output_len 512 still reported plausible throughput; correctness was degraded silently. The scheduler already emitted the needed signal; the worker simply did not consult it.

5.4 CUTLASS schedule/epilogue pairing

Rule. Mainloop schedule and epilogue schedule must match. A residual-fused kernel with `TmaWarpSpecializedCooperative` epilogue requires a cooperative mainloop schedule.

Enforcement. The v3 catalog uses a CUDA `static_assert` on the paired schedule types at template instantiation; the variant table generator writes Rust and CUDA sides together, so adding a mismatched variant fails compilation. In v2, the o_proj residual dispatch pins to the hand-verified variant `v1` ($128 \times 128 \times 128$ Coop/Coop), and the catalog has been audited for pairings—the offending variants are cutlass_fp8_gemm_v$\{0, 2, 3, 4, 6, 8, 11, 12, 13, 14\}$ and cutlass_fp8_gemm_residual_v$\{0, 2, 4, 5, 6\}$.

5.5 No `unwrap()` or `String` errors across crates

Rule. All errors crossing crate boundaries are `Result<T, RvllmError>`. The error variants carry structured context—stream identifier, kernel name, launch configuration, GEMM shape, workspace size needed vs given—not a stringified `DriverError`.

Failure mode. During the April 15 debugging pass, several downstream error paths presented the root-cause `cuGraphLaunch failed` as the opaque string `"dtoh event sync: DriverError(CUDA_ERROR_I ...)"`. The failing kernel name, stream identifier, bucket size, and kernel-launch arguments were discarded at the first `map_err`. Recovery required `compute-sanitizer` to re-derive what the error path should have carried.

6 Measured recovery, April 16, 2026

Three root-cause fixes, in three commits, took the engine from 9,531 tok/s to 19,287 tok/s at $N = 128$ in one working day. None were stopgaps; each eliminated a structural failure mode documented in Section 5.

Commit	**Fix**	**Δ**(N=128)
`6dabb76a5`	Track `last_padded_batch`; only patch metadata when bucket matches	crash → runs
`31716269d`	Re-enable fused FP8 o_proj + residual on variant `v1` (Coop/Coop)	+83%
`eb9e247fd`	Real block-change detection via `block_table_update` (not CoW-only)	+10%

In addition, `libfa3_kernels.so` was built on the deploy host (23 MB, once), eliminating a silent `.ptx` attention fallback path. All four changes taken together restore the engine to a configuration where every decode step uses the intended kernels, with no silent degradation in the dispatch.

7 Results

7.1 Experimental setup

All measurements in this section are on a single NVIDIA H100 SXM 80 GB (driver compatible with CUDA 12.4), running Qwen2.5-7B from the public HuggingFace checkpoint `Qwen/Qwen2.5-7B`, quantized to FP8 E4M3 at engine startup with per-tensor scale. The benchmark harness is `rvllm-v2-bench` (the `crates/rvllm-v2/src/bin/bench.rs` binary), which drives the engine directly without the HTTP layer, runs 3 iterations per batch size after 1 warmup, and reports tokens per second as (batch × output_len)/elapsed. Greedy decoding (`temperature`=0, `ignore_eos`=true), 512 output tokens per request, CUDA graphs enabled, all `policy.json` entries pre-populated by `autotune-cutlass`, `libfa3_kernels.so` built from the repository's build script. SHA: `eb9e247fd`. Date: April 16, 2026.

7.2 Throughput by batch size

N	tok/s	stdev
1	195.2	0.0
4	770.2	0.7
8	1,529.6	0.8
16	3,011.8	0.6
32	5,935.0	0.4
64	11,064.3	1.6
128	**19,287.5**	38.5

7.3 Where the cycles go

Per-kernel time (nsys profile, $N = 32$, mid-decode), from an earlier characterization of the same model on the same hardware:

- CUTLASS FP8 GEMM (fused residual, o_proj): 15.9–65.6 μs depending on bucket and variant.
- FA3 SM90 paged decode: 7.5 μs per layer at $N = 32$.
- fused RMSNorm + FP8 quant (per-token): 6.0–7.3 μs.
- fused SiLU*mul + FP8 quant: 14.3 μs.

- cuBLAS LM-head HGEMM: $\sim$483 μs on $(32, 152064, 3584)$.

The LM-head HGEMM dominates a decode step at $N = 32$. CUTLASS FP8 alternatives for this shape were measured at $\sim$3% slower than cuBLAS at $M = 128$ and are disabled in dispatch.

8 Related work

vLLM [1] introduced the paged-KV cache and continuous batching patterns that rvLLM's scheduler implements. The paged KV layout, block-manager structure, and preemption semantics follow vLLM's conventions closely; the departure is the Rust runtime, the graph-replay discipline, and the explicit rejection of the dispatch fallback chain.

FlashAttention-3 [?] supplies the SM90 paged-decode kernel; rvLLM links directly against a `.so` built from the Hopper source tree and does not re-implement the attention kernel. The `.so` build, variant selection, and workspace contract are rvLLM's responsibility.

CUTLASS [?] supplies the FP8 GEMM templates. rvLLM generates the variant catalog, enforces the schedule-pairing rule, and ships a per-shape autotune policy as a build artifact; the CUTLASS library itself does not enforce pairings or manage workspaces across a runtime.

TensorRT-LLM [?] and SGLang [?] occupy the same space but both assume a CUDA-Python host process. rvLLM's aim is to demonstrate that a GPU-serving engine can be written entirely in a systems language with no Python interpreter in the critical path, and that doing so exposes the kernel-level correctness invariants that a CUDA-Python host can paper over.

9 Limitations and future work

The engine at `eb9e247fd` is single-GPU only. Multi-GPU tensor parallelism exists in a sibling crate but is not wired into the v2 FP8 path. Pipeline parallelism is not implemented. The model catalog validated end-to-end on this FP8 path is Qwen2 family, Llama 3 family, Mistral 7B, and Gemma 2; other supported architectures compile but have not been validated against HuggingFace reference on this version. Speculative decode, LoRA serving, and tool-calling guided decoding are present in the Python-free API layer but outside this paper.

The v3 rewrite consolidates the engine into 16 crates with a DAG-enforced dependency graph and per-crate LoC budgets, and makes each of the five invariants in Section 5 either a compile error, a type-level rejection, or a single-source-of-truth runtime check. The CUDA `static_assert` on schedule pairings, the removal of the second metadata upload path, the `GraphSafe` marker trait rejecting `&mut HbmArena` inside `CaptureScope` closures, and the SHA-pinned `policy.json` artifact each eliminate one of the three bug classes documented above. v3 is tracked under `v3/SPEC.md` and `v3/IMPL_PLAN.md` in the repository.

10 Conclusion

rvLLM demonstrates that an LLM inference engine competitive with modern Python-hosted serving stacks can be written entirely in Rust, with a single compute stream per worker, a frozen metadata layout per graph bucket, and a kernel catalog in which every variant's workspace and schedule are known at engine initialization. The engine achieves 19,287 tok/s serving Qwen2.5-7B in FP8 on a single H100, with 311 kernel launches per decode step compressed into one `cuGraphLaunch` replay and a total of one DtoH and one HtoD copy per step on the hot path.

The path to this number was not always direct: three failure modes documented in Section 5 produced either silent correctness bugs or `CUDA_ERROR_ILLEGAL_ADDRESS` only under graph

replay, and the engine spent one April day at roughly half the peak throughput because of exactly these issues. The fixes, three commits in one day, recovered the full performance envelope without introducing fallbacks. The v3 rewrite makes each of the three bug classes unrepresentable in the type system or caught at compile time.

References

[1] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023. doi: 10.1145/3600006.3613165.